ADEMP pre-registration: 05-nof1-design-sensitivity

2026-05-07 16:45 PDT

This document pre-registers the simulation programme that underlies `analysis/report/05-nof1-design-sensitivity/`. ADEMP structure follows Morris, White and Crowther (2019).

Common infrastructure

Trial design. Hendrickson hybrid OL+BDC N-of-1 design as the primary aggregated variant, with the alternating A/B/A/B N-of-1 variant retained as a sensitivity comparator at sweeps that vary cycle count or period length. Two-period crossover and parallel-group RCT included as cross-design comparators in sweeps S6, S7, S8.

Sample size. $N = 35$ participants per design (Hendrickson 2020 minimum; per-participant parity with the parallel RCT comparator), unless explicitly varied by the sweep.

Drug context. Prazosin for PTSD nightmares; effect size -2.0 nightmares/week at the baseline cell.

Carryover. Exponential decay with half-life $t_{1/2} = 3$ days at the baseline cell.

Replicates. $n_{\text{reps}} = 500$ per cell for sweeps S1-S11; $n_{\text{reps}} = 2000$ per cell for S12 (null calibration).

Random-number control. Per-replicate seed derived from cell descriptor hash and base seed; recorded in `01-base-seed.txt`.

Primary estimand. The treatment main-effect coefficient β_D in the linear mixed-effects analysis-model formula $\text{Sx} \sim \text{Dbc} + t + (1 \mid \text{ptID})$ with `corCAR1(form = ~t|ptID)`. The biomarker-by-treatment interaction $\beta_{bm:D}$ is a secondary estimand in sweep S10 only.

Performance measures. Power $\pi = \Pr(p < 0.05)$ for the two-sided $H_0 : \beta_D = 0$, type I error rate under $\beta_D = 0$, bias of $\hat{\beta}_D$, empirical Monte Carlo SE of $\hat{\beta}_D$ across replicates, mean within-replicate model SE, nominal-95% CI coverage, and convergence rate. Each reported with the binomial-proportion MCSE $\sqrt{\hat{\pi}(1-\hat{\pi})/n_{\text{reps}}}$ for proportions and the Monte Carlo SE of the within-replicate mean estimator for continuous measures.

Sweeps S1-S12

Per the manuscript's Table 1 (Section 3 'Simulation design'):

Sweep	Axis	Cells	n_{reps}
S1	Effect size	$\beta_D \in \{-3, -2.5, -2, -1.5, -1, -0.5, 0\}$	500
S2	Carryover half-life	$t_{1/2} \in \{0.5, 1, 3, 5, 7\}$ days	500
S3	Decay-form misspecification	linear / exponential / Weibull	500
S4	Between-patient SD	$\tau \in \{0, 0.5, 1, 1.5, 2\}$	500
S5	Within-patient SD	$\sigma \in \{0.5, 1, 1.5, 2\}$	500
S6	Cycles per patient	$k \in \{1, 2, 3, 4, 6\}$	500
S7	Total participant count	$N \in \{15, 25, 35, 50, 70, 100\}$	500
S8	AR(1) serial correlation	$\rho \in \{0, 0.15, 0.3, 0.45, 0.6\}$	500
S9	Period length	$L \in \{1, 2, 3, 4, 6\}$ weeks	500
S10	Biomarker c_{bm} × DGP architecture	$\{0, 0.3, 0.6\} \times \{A, B\}$	500

Sweep	Axis	Cells	n_{reps}
S11	Carryover-by-AR(1) misspecification	DGP $t_{1/2}$ × analysis $t_{1/2}$, 4×4	500
S12	High-replicate null calibration	$\beta_D = 0$, single cell	2000

For each sweep, the **Aims** are to characterise the sensitivity of the primary estimand to the perturbed axis; **Data-generating mechanism** is the baseline DGP with the named axis varied; **Estimands** and **Methods** are inherited from the common infrastructure above; **Performance measures** are inherited and supplemented by the sweep-specific visualisation (power-by-axis plot with MCSE error bars).

Deliverables

- Driver scripts at `analysis/scripts/nof1-design-sensitivity/`.
- `02-grid-summary.rds` and `02-sensitivity-summary.rds` checkpointed under `analysis/data/`.
- ADEMP results table per sweep with MCSE columns alongside every estimate.
- Figures plotting power and bias as a function of each axis with MCSE error bars.

Deviations log

`02-deviations-log.md` records any departure from the cells, estimands, methods, or performance measures pre-registered above.

Source: `~/prj/alz/10-pmsimstats-ng/pmsimstats-ng/analysis/scripts/nof1-design-sensitivity/00-ademp-pre-registration.md`